

WASP-10b

R-0890

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-10_180826 · created 2026-07-29T01:55:09+00:00

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2458356.7607 +/- 0.004 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.135 +/- 0.04
Transit depth (Rp/Rs)^2	1.83 +/- 1.09 [%]
Orbital Inclination (inc)	89.32 +/- 1.48
Airmass coefficient 1 (a1)	0.938 +/- 0.04
Airmass coefficient 2 (a2)	0.056 +/- 0.024
Scatter in the residuals of the lightcurve fit is	4.22 %
Best Comparison Star	#2 - [214, 327]
Optimal Method	PSF photometry
Transit Duration (day)	0.0896 +/- 0.0065

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R* fit vs published	0.135 ± 0.04 vs 0.1592 (deviation 0.6σ)
Duration fit vs published	0.0896 d vs 0.0946 d (deviation 0.77σ)
Mid-time O–C	7.7 min
Depth SNR	3.4
Residual scatter	4.22 %

Light curve

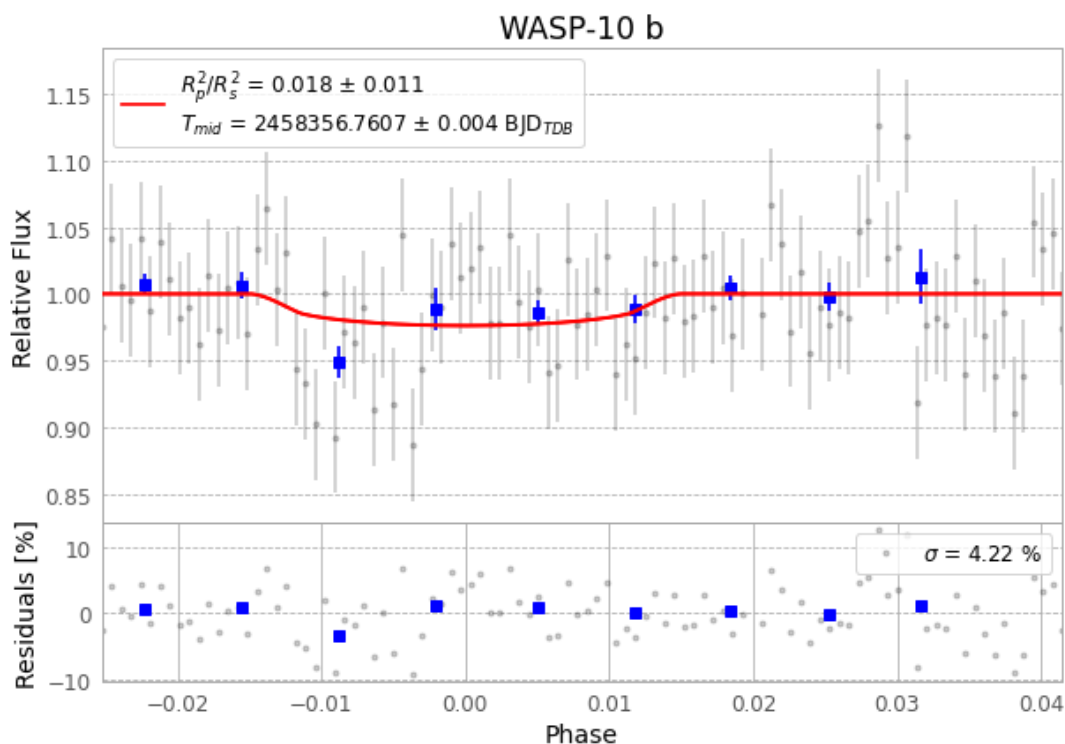


Figure 1 — FinalLightCurve_WASP-10 b_2018-08-25.png (SHA-256 b5a615d5d7747f56...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [364, 173], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 efec15804dd39413...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ead60b9

Data provenance

99 FITS frames (65 MB), WASP-10180826041512.FITS → WASP-10180826091212.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-10-180826.log (c93b6a20c15e...), FinalLightCurve_WASP-10 b_2018-08-25.png (b5a615d5d774...), FinalLightCurve_WASP-10 b_2018-08-25.csv (b395c7157260...)

Compute resources

Wall time 125.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-29 01:55Z.